



Extremal Problems of the Density for Vibrating String Equations with Applications to Gap and Ratio of Eigenvalues

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Abstract

The present paper obtains the infimum of the densities for vibrating string equations in terms of the gap and ratio of the first two eigenvalues. The main result in this paper can be viewed as a new version of the Lyapunov inequality involving the first two eigenvalues. Some new estimates of the gap and ratio are obtained. Furthermore, the classical Lyapunov inequality can be deduced by the main conclusion.

Keywords Vibrating string equation · Density · Eigenvalue gap (ratio) · Extremal problem · Lyapunov inequality

Mathematics Subject Classification Primary 34B20; Secondary 34L05

1 Introduction

In this paper we consider the eigenvalue problem of the vibrating string equation

$$-y'' = \lambda wy, \quad y = y(x), \quad x \in (0, 1) \quad (1.1)$$

associated to the Dirichlet boundary condition

$$y(0) = 0 = y(1), \quad (1.2)$$

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where the density w is a nonnegative integrable function on $[0, 1]$. The natural frequencies of vibration of the string are determined by the eigenvalues of (1.1) and (1.2). The eigenvalue problem of (1.1) and (1.2) is the special case of the general Sturm–Liouville eigenvalue problem

$$-(py')' + qy = \lambda wy, \quad y = y(x), \quad x \in (0, 1), \quad y(0) = 0 = y(1), \quad (1.3)$$

where

$$1/p, \quad q, \quad w \in L^1[0, 1], \quad \|w\|_1 > 0, \quad \text{and} \quad p(x) > 0, \quad w(x) \geq 0 \text{ a.e. } x \in [0, 1]. \quad (1.4)$$

Here $L^1[0, 1]$ denotes the integrable functions on $[0, 1]$, with the norm

$$\|w\|_1 := \int_0^1 |w(x)| dx.$$

By the classical spectral theory of problem (1.3), cf. [8], it is known that the eigenvalue problem (1.1) and (1.2) has only discrete and real eigenvalues. Denote $\lambda_n(w)$ the n th eigenvalue of (1.1) and (1.2). Then

$$0 < \lambda_1(w) < \cdots < \lambda_n(w) < \cdots, \quad \lambda_n(w) \rightarrow \infty, \quad n \rightarrow \infty. \quad (1.5)$$

Let $0 < \lambda_1 < \lambda_2 < \cdots < \lambda_m$ be fixed, where $m \geq 1$ is an integer. Define

$$\Omega(\lambda_1, \dots, \lambda_m) := \{w \in L^1[0, 1] : \|w\|_1 > 0, \quad \lambda_j(w) = \lambda_j, \quad 1 \leq j \leq m, \\ \text{and } w(1-x) = w(x) \geq 0, \text{ a.e. } x \in [0, 1]\}, \quad (1.6)$$

here we call w is *symmetric* on $[0, 1]$ if $w(1-x) = w(x)$, a.e. $x \in [0, 1]$. We are interested in the infimum of the norm $\|w\|_1$ of the densities w in $\Omega(\lambda_1, \dots, \lambda_m)$. Therefore we study the quantity

$$E(\lambda_1, \dots, \lambda_m) := \inf\{\|w\|_1 : w \in \Omega(\lambda_1, \dots, \lambda_m)\}. \quad (1.7)$$

For the simple case $m = 1$, the classical Lyapunov inequality(cf. [14], [19, (1.2)], [2, p.177]) gives that

$$E(\lambda_1) = \frac{4}{\lambda_1}. \quad (1.8)$$

Furthermore, from [17, Theorem 2.1], we know for any $w \in \Omega(\lambda_1)$, $\|w\|_1 > 4/\lambda_1$. Hence, the infimum $4/\lambda_1$ is not achieved in $\Omega(\lambda_1)$, however it can be obtained if

$$w = \frac{4}{\lambda_1} \delta_{1/2},$$

where $\delta_{1/2}$ is the Dirac delta distribution at $1/2$, see the detail in (2.1) of Sect. 2.

But what about the quantity of $E(\lambda_1, \dots, \lambda_m)$ for $m \geq 2$? This question is not easy to answer directly. In the present paper we are mainly concerned with the case $m = 2$.

Applying the spectral theory of differential equations with distribution densities, we prove that

$$E(\lambda_1, \lambda_2) = \frac{4\lambda_2}{\lambda_1(\lambda_2 - \lambda_1)} \quad (1.9)$$

if $\lambda_2 \geq 2\lambda_1$, see Theorem 3.1.

In fact, the quantity $E(\lambda_1, \lambda_2)$ implies that for any arbitrary symmetric density $w \in L^1[0, 1]$ with $\|w\|_1 \neq 0$, if $\lambda_2 \geq 2\lambda_1$, then (see corollary 3.4)

$$(\lambda_2 - \lambda_1) \int_0^1 w \geq \frac{4\lambda_2}{\lambda_1}. \quad (1.10)$$

This inequality gives the relation between the gap and ratio of the first two eigenvalues, which is a new version of the classical Lyapunov inequality.

There are many previous results on estimates of the gap (cf. [11, Theorem 2.1], [13,18]) and ratio (cf. [1, Theorem 5.1], [5,6]) of the first two eigenvalues. But few results are on the inequalities involving both the gap and ratio. Unfortunately, the validity of the (1.10) needs the extra condition $\lambda_2 \geq 2\lambda_1$. We conjecture that the inequality (1.10) is always true for symmetric weight without the extra condition.

Clearly, a lower bound of the gap of the first two eigenvalues

$$\lambda_2(w) - \lambda_1(w) \geq 4 \quad (1.11)$$

can be deduced from (1.10) if $\int_0^1 w = 1$. But we point out that the above inequality is always true without the restriction $\lambda_2(w) \geq 2\lambda_1(w)$. This conclusion can be deduced from the main results in the recent paper [4, Theorem 3.1]. If the weight w is symmetric, we show that 4 is indeed the infimum of the gap of the first two eigenvalues, see Corollary 2.3.

We also indicate that there exists many cases such that the condition $\lambda_2(w) \geq 2\lambda_1(w)$ is satisfied, and thus some new estimates of the gap can be obtained by using (1.10), see Corollary 3.8.

Following this section, the lower bound of the gap is obtained in Sect. 2, see Corollaries 2.3 and 2.5. The equality (1.9) is proved in Theorem 3.1 of Sect. 3 by using Mercer Theorem.

2 The Gap of the First Two Eigenvalues

In this section, we study the extremal problem of the gap of the first two eigenvalues of problem (1.1) and (1.2). Firstly, we introduce the continuity of the eigenvalues of (1.1) and (1.2) with respect to the density in weak convergence. To this end, we briefly review some basic facts about the Dirac delta distribution and weak convergence in the following.

For a real function $\mu : [0, 1] \rightarrow \mathbb{R}$, the *total variation* of μ (on $[0, 1]$) is defined as

$$\|\mu\|_v := \sup \left\{ \sum_{i=0}^{n-1} |\mu(x_{i+1}) - \mu(x_i)| : 0 = x_0 < x_1 < \cdots < x_{n-1} < x_n = 1, n \in \mathbb{N} \right\}.$$

Define a space of measures on $[0, 1]$ by

$$\mathcal{M}_0 := \{\mu : [0, 1] \rightarrow \mathbb{R} : \mu(0) = 0, \mu(x+) = \mu(x) \forall x \in (0, 1), \|\mu\|_v < +\infty\}.$$

Here, for $x \in (0, 1)$, $\mu(x+) := \lim_{s \downarrow x} \mu(s)$ is the right-limit. By the Riesz representation theorem (cf., [15]), $(\mathcal{M}_0, \|\cdot\|_v)$ is the same as the dual space $C[0, 1]^*$, where the space $C[0, 1]$ denote the Banach space of continuous functions on $[0, 1]$ with the supremum norm. In fact, for any $\mu \in \mathcal{M}_0$, using the Riemann–Stieltjes integrals, we can define $\mu^* \in C[0, 1]^*$ by

$$\mu^*(u) = \int_0^1 u(x) d\mu(x), \text{ for any } u \in C[0, 1].$$

We also denote $d\mu$ as $\rho = d\mu$, and called it “distribution density”. Clearly, if $w \in L^1[0, 1]$, w is a distribution density.

As an important example of distribution density, the Dirac delta distribution δ at a point $a \in (0, 1)$ is defined by

$$\int_0^1 \delta_a(x) u(x) dx = u(a), \text{ for any } u \in C[0, 1]. \quad (2.1)$$

In fact, it can be viewed as the distribution density of the Heaviside function,

$$H_a(x) = \begin{cases} 0, & x \in [0, a), \\ 1, & x \in [a, 1]. \end{cases}$$

Moreover, we say that $w_n \in L^1[0, 1]$ is weak star convergent to the distribution density ρ , if for any $u \in C[0, 1]$,

$$\lim_{n \rightarrow \infty} \int_0^1 u w_n = \int_0^1 u \rho, \text{ written as } w_n \xrightarrow{w} \rho.$$

For more details about the above weak convergence and weak topologies, one can refer to [7, 15]. For simplicity, this weak star convergence is also called the weak convergence in the present paper.

With this notation we will consider (1.1) and (1.2) with distribution density for two special cases: the distribution density ρ_1 of one point and the symmetric distribution density ρ_2 of two points, where

$$\begin{aligned} \rho_1(x) &= r\delta_a(x), \quad r > 0, \quad 0 < a < 1, \\ \rho_2(x) &= r\delta_a(x) + r\delta_{1-a}(x), \quad r > 0, \quad 0 < a < 1/2. \end{aligned} \quad (2.2)$$

From [20, Lemma 3.3], we know that the problem (1.1) and (1.2) with $w = \rho_1$ has exactly one eigenvalue

$$\lambda_1(\rho_1) = \frac{1}{ra(1-a)} \quad (2.3)$$

and the problem (1.1) and (1.2) with $w = \rho_2$ has exactly two simple eigenvalues

$$\lambda_1(\rho_2) = \frac{2}{ra}, \quad \lambda_2(\rho_2) = \frac{2}{ra(1-2a)}. \quad (2.4)$$

Furthermore, the following result gives the continuity of eigenvalues on density with respect to the weak topology.

Proposition 2.1 (cf. [20, Theorem 4.1(ii)]) *Let $\lambda_k(w)$ be the k th eigenvalue of (1.1) and (1.2), $\lambda_k(\rho_j)$ be defined as in (2.3) and (2.4) for $j = 1, 2$.*

(i) *If $w_n \in L^1[0, 1]$, $n \geq 1$ and $w_n \xrightarrow{w} \rho_1$, then $\lambda_1(w_n) \rightarrow \lambda_1(\rho_1)$ and $\lambda_k(w_n) \rightarrow \infty$ for $k \geq 2$, as $n \rightarrow \infty$.*

(ii) *If $w_n \in L^1[0, 1]$, $n \geq 1$ and $w_n \xrightarrow{w} \rho_2$, then $\lambda_k(w_n) \rightarrow \lambda_k(\rho_2)$ for $k = 1, 2$ and $\lambda_k(w_n) \rightarrow \infty$ for $k \geq 3$, as $n \rightarrow \infty$.*

The main result of this section gives the lower bound of $E(\lambda_1, \lambda_2)$ for the case $k = 2$ involving only the gap of the first two eigenvalues, where the definitions of $\Omega(\lambda_1, \dots, \lambda_k)$ and $E(\lambda_1, \dots, \lambda_k)$ are in (1.6) and (1.7), respectively.

Theorem 2.2 *Let $E(\lambda_1, \lambda_2)$ be defined as in (1.7) for $k = 2$. Then*

$$E(\lambda_1, \lambda_2) \geq \frac{4}{\lambda_2 - \lambda_1}. \quad (2.5)$$

Proof Let $\phi_1(x)$ and $\phi_2(x)$ be the corresponding eigenfunctions of λ_1 and λ_2 , respectively, satisfying $\phi_1'(0) = \phi_2'(0) = 1$. Then $\phi_1(x) > 0$ on $(0, 1)$ and $\phi_1'(1/2) = \phi_2'(1/2) = 0$ by the symmetry of w . Set

$$u(x) = \frac{\phi_2(x)}{\phi_1(x)}, \quad x \in (0, 1).$$

Then

$$u(0) = \lim_{x \rightarrow 0} \frac{\phi_2(x)}{\phi_1(x)} = \frac{\phi_2'(0)}{\phi_1'(0)}$$

and

$$-(\phi_1^2 u')' = \Delta \phi_1^2 w u, \quad x \in (0, 1), \quad \phi_1^2 u'(0) = 0, \quad u(1/2) = 0,$$

where $\Delta = \lambda_2 - \lambda_1$. Integrating two sides of the above equation on $[0, x]$ gives

$$\phi_1^2 u'(x) = -\Delta \int_0^x \phi_1^2(t) u(t) w(t) dt,$$

and hence

$$u'(x) = -\Delta \frac{1}{\phi_1^2(x)} \int_0^x \phi_1^2(t) u(t) w(t) dt, \quad x \in (0, 1). \quad (2.6)$$

Note that $u(x) > 0$ on $(0, 1/2)$. Then (2.6) means that $u'(x) \leq 0$ on $(0, 1/2)$. Integrating two sides of (2.6) on $[0, 1/2]$ yields that

$$u(0) = \Delta \int_0^{1/2} \int_0^x \frac{\phi_1^2(t)}{\phi_1^2(x)} w(t) u(t) dt dx. \quad (2.7)$$

Note that ϕ_1 is the first eigenfunction,

$$-\phi_1''(x) = \lambda_1 w(x) \phi_1(x), \quad x \in (0, 1), \quad \phi_1(0) = 0 = \phi_1(1),$$

$\phi_1'(1/2) = 0$ and $\phi_1(x) > 0$, $x \in (0, 1)$. Then

$$\phi_1'(x) = \lambda_1 \int_x^{1/2} \phi_1(x) w(x) dx \geq 0, \quad x \in [0, 1/2]$$

since $\lambda_1 > 0$. This means that $\phi_1(x)$ or $\phi_1^2(x)$ is increasing on $(0, 1/2)$. Then

$$\phi_1^2(x) \geq \phi_1^2(t), \quad \frac{1}{2} \geq x \geq t \geq 0, \quad \text{or} \quad \frac{\phi_1^2(t)}{\phi_1^2(x)} \leq 1, \quad \frac{1}{2} \geq x \geq t \geq 0.$$

This together with (2.7) and $u'(x) < 0$ on $(0, 1/2)$ leads us to

$$\begin{aligned} u(0) &\leq \Delta \int_0^{1/2} \int_0^x w(t) u(t) dt dx = \Delta \int_0^{1/2} w(t) u(t) (1/2 - t) dt \\ &< \frac{1}{2} u(0) \Delta \int_0^{1/2} w(t) dt. \end{aligned} \quad (2.8)$$

The last strict inequality follows from

$$\int_0^{1/2} x w(x) dx > 0,$$

for otherwise we have $w(x) \equiv 0$ a.e. $x \in [0, 1]$. Then $\Delta \int_0^{1/2} w(x) dx > 2$, and hence $\Delta \int_0^1 w(x) dx > 4$, or $E(\lambda_1, \lambda_2) \geq 4/(\lambda_2 - \lambda_1)$ since w is symmetric. $\square$

We remark that for fixed λ_1, λ_2 , the value $\frac{4}{\lambda_2 - \lambda_1}$ is not the extremum, i.e., $E(\lambda_1, \lambda_2) \neq \frac{4}{\lambda_2 - \lambda_1}$ for some cases, see the results in Theorem 3.1. However, from the results of Theorem 2.2, we have the following estimates of the gap of the first two eigenvalues.

Corollary 2.3 *Let λ_1 and λ_2 be the first and second eigenvalues of (1.1) and (1.2), respectively. Suppose that $w(x)$ is symmetric on $[0, 1]$ and $\int_0^1 w(x) dx = 1$. Then*

$$\lambda_2 - \lambda_1 \geq 4,$$

and 4 is the best constant.

Proof We need only to prove that there exists a series of symmetric w_n with $\int_0^1 w_n(x)dx = 1$ such that $\lambda_2(w_n) - \lambda_1(w_n) \rightarrow 4$ as $n \rightarrow \infty$. Choose the distribution density just as (2.2),

$$\rho_\epsilon(x) = \frac{1}{2}[\delta_\epsilon(x) + \delta_{1-\epsilon}(x)], 0 < \epsilon < 1/2. \quad (2.9)$$

Then from (2.4) one sees that

$$\lambda_1(\rho_\epsilon) = \frac{2}{\epsilon}, \quad \lambda_2(\rho_\epsilon) = \frac{2}{\epsilon(1-2\epsilon)}.$$

Hence

$$\lambda_2(\rho_\epsilon) - \lambda_1(\rho_\epsilon) = \frac{4}{1-2\epsilon} \rightarrow 4, \quad \epsilon \rightarrow 0. \quad (2.10)$$

For fixed $\epsilon > 0$, choose $w_n(\cdot, \epsilon) \in L^1[0, 1]$ for large enough n defined by

$$w_n(x, \epsilon) := \begin{cases} n, & x \in \left(\epsilon - \frac{1}{4n}, \epsilon + \frac{1}{4n}\right) \cup \left(1 - \epsilon - \frac{1}{4n}, 1 - \epsilon + \frac{1}{4n}\right), \\ 0, & \text{otherwise.} \end{cases} \quad (2.11)$$

It is easy to verify that the series of $w_n(\cdot, \epsilon)$ is symmetric, $\int_0^1 w_n(x, \epsilon)dx = 1$ and $w_n(\cdot, \epsilon) \xrightarrow{w} \rho_\epsilon$ as $n \rightarrow \infty$. It follows from Proposition 2.1 that there exists a series of symmetric w_n satisfying, $\int_0^1 w_n(x)dx = 1$ and

$$\lambda_2(w_n) - \lambda_1(w_n) \rightarrow 4 \text{ as } n \rightarrow \infty.$$

The proof is finished. $\square$

In the recent paper [4, Theorem 3.1], the authors showed that if $w(x)$ is a symmetric continuous function, then

$$\lambda_2 - \lambda_1 > \max \left\{ \frac{1}{\int_0^{1/2} (1/2 - x)w(x)dx}, \frac{\pi^2}{w_M} \right\},$$

where $w_M := \max\{w(x) : x \in [0, 1]\}$. Of course, if $\int_0^1 w(x)dx = 1$, then

$$\int_0^{\frac{1}{2}} \left(\frac{1}{2} - x\right) w(x)dx < \int_0^{\frac{1}{2}} \frac{1}{2} w(x)dx = \frac{1}{4}.$$

Hence it can be deduced from the above two inequalities that the lower bound of $\lambda_2 - \lambda_1$ is 4.

Corollary 2.3 indicates that 4 is the best constant in the gap inequality if w is symmetric. In fact, for some non-symmetric density the gap inequality still holds, see Corollary 2.5. To prove this result, we need Mercer's theorem (cf. [3,16]) for

the Sturm–Liouville eigenvalue problem (1.3). See Lemma 2.4. For self-containment, we give a short proof here. Furthermore, we will apply Mercer’s theorem to prove Theorem 3.1 in Sect. 3.

Lemma 2.4 (Mercer’s theorem for Sturm–Liouville problems) *Let w, p, q be defined as in (1.4) and λ_n the n th eigenvalue of (1.3). For $\lambda_0 \neq \lambda_n, n \geq 1$, let $G(x, t)$ be the Green function of the problem (1.3) at λ_0 . Then for $n \geq 1$,*

$$\int_0^1 \cdots \int_0^1 G(s_1, s_2) \cdots G(s_n, s_1) w(s_1) \cdots w(s_n) ds_1 \cdots ds_n = \sum_{j=1}^{\infty} \frac{1}{(\lambda_j - \lambda_0)^n}. \quad (2.12)$$

Proof By the definition of the Green function $G(x, t)$, the eigenvalue problem (1.3) is equivalent to the following integral equation:

$$\int_0^1 G(x, t) w(t) \phi(t) dt = \frac{1}{\lambda - \lambda_0} \phi(x), \quad (2.13)$$

where λ_0 is not an eigenvalue of (1.3).

Suppose that $\{\phi_j, j = 1, 2, \dots\}$ is the orthogonal normalized eigenfunction system of (1.3) or (2.13). By [3, §3.5.1, (52), p.134], the kernel $G(x, t)$ could be expanded in a series

$$G(s, t) = \sum_{j=1}^{\infty} \frac{1}{\lambda_j - \lambda_0} \phi_j(s) \phi_j(t), \quad (2.14)$$

which would converge absolutely and uniformly in each variable. By the orthogonality of $\{\phi_j, j = 1, 2, \dots\}$ and the uniform convergence of series (2.14), we can get

$$\int_0^1 G(s_1, s_1) w(s_1) ds_1 = \sum_{j=1}^{\infty} \frac{1}{\lambda_j - \lambda_0}.$$

Using the same method, we can prove

$$\int_0^1 \int_0^1 G(s_1, s_2) G(s_2, s_1) w(s_1) w(s_2) ds_1 ds_2 = \sum_{j=1}^{\infty} \frac{1}{(\lambda_j - \lambda_0)^2}.$$

In the same manner, one can prove that (2.12) holds. $\square$

Consider the problem (1.1) and (1.2). The corresponding Green function at $\lambda = 0$ is given by

$$G(x, t) = \begin{cases} x(1-t), & 0 \leq x \leq t \leq 1, \\ t(1-x), & 1 \geq x \geq t \geq 0. \end{cases}$$

Then from Lemma 2.4 and (1.5), one sees that

$$\int_0^1 x(1-x)w(x)dx = \sum_{j=1}^{\infty} \frac{1}{\lambda_j}. \quad (2.15)$$

Hence

$$\int_0^1 w \geq 4 \int_0^1 x(1-x)w(x)dx = 4 \sum_{j=1}^{\infty} \frac{1}{\lambda_j}.$$

If $\lambda_2 \geq \frac{1+\sqrt{5}}{2}\lambda_1$, one can verify that $1/\lambda_1 + 1/\lambda_2 \geq (\lambda_2 - \lambda_1)^{-1}$, and hence

$$\int_0^1 w \geq 4 \sum_{j=1}^{\infty} \frac{1}{\lambda_j} \geq \frac{4}{\lambda_2 - \lambda_1}.$$

In fact we have proven that

Corollary 2.5 *Let λ_n be the n th eigenvalue of (1.1) and (1.2). If $\lambda_2 \geq \frac{1+\sqrt{5}}{2}\lambda_1$, then*

$$\int_0^1 w \geq \frac{4}{\lambda_2 - \lambda_1}.$$

Epecially, if $\int_0^1 w(x)dx = 1$, then $\lambda_2 - \lambda_1 \geq 4$.

Comparing Theorem 2.2 with Corollary 2.5, one sees that the density w doesn't need to be symmetric in Corollary 2.5. In [4,10], some lower bounds of the gap of first two eigenvalues are given for the case where w is symmetric.

3 The Gap and Ratio of the First Two Eigenvalues

This section gives the quantity $E(\lambda_1, \lambda_2)$ in terms of the gap and ratio.

Theorem 3.1 *Let λ_n be the n th eigenvalue of the problem (1.1) and (1.2). Then*

$$E(\lambda_1, \lambda_2) = \frac{4\lambda_2}{\lambda_1(\lambda_2 - \lambda_1)} \quad (3.1)$$

if one of the following two conditions holds,

$$(i) \lambda_1 \sum_{j=1}^{\infty} \frac{1}{\lambda_{2j-1}} \leq \lambda_2 \sum_{j=1}^{\infty} \frac{1}{\lambda_{2j}}, \quad (ii) \lambda_2 \geq 2\lambda_1. \quad (3.2)$$

To prove Theorem 3.1 we need the following preliminary results. The first one is a consequence of Mercer's theorem. See Lemma 2.4.

Corollary 3.2 *Let λ_n be the n th eigenvalue of (1.1) and (1.2). If $w(x)$ is symmetric on $[0, 1]$. Then it holds that*

$$\int_0^{1/2} xw(x)dx = \sum_{j=1}^{\infty} \frac{1}{\lambda_{2j-1}}, \quad \int_0^{1/2} x(1-2x)w(x)dx = \sum_{j=1}^{\infty} \frac{1}{\lambda_{2j}}. \quad (3.3)$$

Proof Clearly, λ_{2j-1} and λ_{2j} are the j th eigenvalues of

$$-y'' = \lambda wy, \quad y(0) = 0 = y'(1/2) \quad (3.4)$$

and

$$-y'' = \lambda wy, \quad y(0) = 0 = y(1/2), \quad (3.5)$$

respectively, for $j \geq 1$ by the symmetry of w . Note that $\lambda_n > 0$, $n \geq 1$. Then we can choose $\lambda_0 = 0$ in Lemma 2.4. The corresponding Green functions For (3.4) and (3.5) are given respectively by

$$G_1(x, t) = \begin{cases} x, & 0 \leq x \leq t \leq 1/2, \\ t, & 1/2 \geq x \geq t \geq 0, \end{cases} \quad G_2(x, t) = \begin{cases} x(1-2t), & 0 \leq x \leq t \leq 1/2, \\ t(1-2x), & 1/2 \geq x \geq t \geq 0. \end{cases}$$

So $G_1(x, x) = x$ and $G_2(x, x) = x(1-2x)$ on $[0, 1/2]$. Therefore, applying the formula (2.12) on $[0, 1/2]$ for $n = 1$ yields the equalities (3.3). $\square$

Lemma 3.3 *Suppose that $w \in L^1[0, 1]$ is symmetric and $w(x) \geq 0$ a.e. on $[0, 1]$. If for some $A > B > 0$,*

$$\int_0^{1/2} xw(x)dx = A, \quad \int_0^{1/2} x(1-2x)w(x)dx = B. \quad (3.6)$$

Then

$$\int_0^1 w(x)dx \geq \frac{4A^2}{A-B}. \quad (3.7)$$

Proof It follows from (3.6) that

$$\int_0^{1/2} x^2w(x)dx = \int_0^{1/2} \frac{1}{2}[x - x(1-2x)]w(x)dx = (A-B)/2.$$

By Cauchy inequality,

$$\left(\int_0^{1/2} xw(x)dx \right)^2 \leq \int_0^{1/2} w(x)dx \int_0^{1/2} x^2w(x)dx.$$

Hence

$$\int_0^1 w(x)dx = 2 \int_0^{1/2} w(x)dx \geq \frac{4A^2}{A-B}.$$

This completes the proof of Lemma 3.3. $\square$

The proof of Theorem 3.1 It follows from Corollary 3.2 that

$$\int_0^{1/2} xw(x)dx = \sum_{j=1}^{\infty} \frac{1}{\lambda_{2j-1}} =: \sigma_1, \quad \int_0^{1/2} x(1-2x)w(x)dx = \sum_{j=1}^{\infty} \frac{1}{\lambda_{2j}} =: \sigma_2.$$

Clearly, $\sigma_1 > \sigma_2$. Therefore, Lemma 3.3 yields that

$$\int_0^1 w(x)dx \geq \frac{4\sigma_1^2}{\sigma_1 - \sigma_2}.$$

Set $\sigma_1 = t/\lambda_1$ and $\sigma_2 = s/\lambda_2$, then $s, t \in [1, \infty)$ and

$$\int_0^1 w(x)dx \geq \frac{4t^2\lambda_2}{\lambda_1(t\lambda_2 - s\lambda_1)}.$$

We claim that under the assumption in Theorem 3.1,

$$\frac{4t^2\lambda_2}{\lambda_1(t\lambda_2 - s\lambda_1)} \geq \frac{4\lambda_2}{\lambda_1(\lambda_2 - \lambda_1)}.$$

Since $t\lambda_2 = \sigma_1\lambda_1\lambda_2 > \sigma_2\lambda_1\lambda_2 = s\lambda_1$, the above inequality leads to

$$(t^2 - t)\lambda_2 \geq (t^2 - s)\lambda_1.$$

If the condition (i) of (3.2) holds, then $s \geq t$, and hence

$$(t^2 - t)\lambda_2 \geq (t^2 - t)\lambda_1 \geq (t^2 - s)\lambda_1.$$

If (ii) of (3.2) holds, i.e. $\lambda_2 \geq 2\lambda_1$, then

$$(t^2 - t)\lambda_2 \geq 2(t^2 - t)\lambda_1 = (t-1)^2\lambda_1 + (t^2 - s)\lambda_1 + (s-1)\lambda_1 \geq (t^2 - s)\lambda_1.$$

This proves $E(\lambda_1, \lambda_2) \geq \frac{4\lambda_2}{\lambda_1(\lambda_2 - \lambda_1)}$. We are proceeding to prove that

$$E(\lambda_1, \lambda_2) = \frac{4\lambda_2}{\lambda_1(\lambda_2 - \lambda_1)}.$$

For given $0 < \lambda_1 < \lambda_2$, take

$$w_0 = r[\delta_{x_0} + \delta_{1-x_0}], \quad r = \frac{2\lambda_2}{\lambda_1(\lambda_2 - \lambda_1)}, \quad x_0 = \frac{\lambda_2 - \lambda_1}{2\lambda_2}, \quad (3.8)$$

then $\lambda_1(w_0) = \lambda_1$ and $\lambda_2(w_0) = \lambda_2$ by (2.4). Clearly,

$$\int_0^1 w_0(x) dx = 2r = \frac{4\lambda_2}{\lambda_1(\lambda_2 - \lambda_1)}.$$

Therefore, by Proposition 2.1 [see (2.11)] there exists $w_n \in L^1[0, 1]$ which is symmetric and $\int_0^1 w_n = \int_0^1 w_0$, $n \geq 1$, such that

$$\lim_{n \rightarrow \infty} \frac{4\lambda_2(w_n)}{\lambda_1(w_n)(\lambda_2(w_n) - \lambda_1(w_n))} = \frac{4\lambda_2}{\lambda_1(\lambda_2 - \lambda_1)}.$$

This completes the proof of Theorem 3.1. $\square$

From Theorem 3.1 one immediately gets

Corollary 3.4 *Let λ_1 and λ_2 be the first and second eigenvalues of (1.1) and (1.2). If $w(x)$ is symmetric on $[0, 1]$, $\int_0^1 w(x) dx = 1$ and $\lambda_2 \geq 2\lambda_1$, then it holds that*

$$\lambda_2 - \lambda_1 \geq 4\lambda_2/\lambda_1. \quad (3.9)$$

Remark 1 It seems to us that the inequality (3.9) is new about the gap and ratio of the first two eigenvalues. Unfortunately, the validity of the (3.9) proved by using the method in the present paper needs the extra condition $\lambda_2 \geq 2\lambda_1$. We hope that (3.9) is always true without this extra condition.

Additionally, under the assumption of Theorem 3.1, the classical Lyapunov inequality is clearly a consequence of Theorem 3.1, and Theorem 2.2 can also be deduced by this theorem either. Furthermore, the classical Lyapunov inequality is the case of limitation as $\lambda_2 \rightarrow \infty$ in Theorem 3.1 which occurs as w weakly converges to a multiplication of the Dirac distribution $\delta_{1/2}$ by Proposition 2.1.

In the sequel, we give some examples that can ensure $\lambda_2 \geq 2\lambda_1$.

Lemma 3.5 *If w is symmetric and $w(x) = 0$, $x \in (0, \delta)$, $0 < \delta < 1/2$, then*

$$\frac{\lambda_2}{\lambda_1} \geq \frac{1}{1 - 2\delta}.$$

Especially, $\lambda_2 \geq 2\lambda_1$ if $\delta = 1/4$.

Proof Let ϕ_j be the j th eigenfunction of (1.1) and (1.2) satisfying $\phi'_j(0) = 1$, $j = 1, 2$. Then

$$\phi_1(\delta) = \delta = \phi_2(\delta), \quad \phi'_1(\delta) = 1 = \phi'_2(\delta), \quad \phi'_1(1/2) = \phi'_2(1/2) = 0. \quad (3.10)$$

Note that the concaveness of $\phi_2(x)$ on $[\delta, 1/2]$ gives

$$\phi_2'(1/2) \leq \frac{\phi_2(1/2) - \phi_2(\delta)}{1/2 - \delta} = \frac{-\delta}{1/2 - \delta}. \quad (3.11)$$

Since $\phi_2(x)/\phi_1(x)$ is non-increasing on $(0, 1/2)$ [see (2.6)] and $\phi_1(\delta)/\phi_1(\delta) = 1$, we get

$$\phi_1(x) > \phi_2(x) > 0 \text{ and } -\phi_1''(x) > -\phi_2''(x), \quad x \in (\delta, 1/2].$$

Integrating $-\phi_1''$ over the interval $[\delta, 1/2]$ we get from (3.10) and (3.11) that

$$\begin{aligned} 1 &= \int_{\delta}^{1/2} -\phi_1'' = \lambda_1 \int_{\delta}^{1/2} w \phi_1 > \lambda_1 \int_{\delta}^{1/2} w \phi_2 \\ &= \frac{\lambda_1}{\lambda_2} \lambda_2 \int_{\delta}^{1/2} w \phi_2 = \frac{\lambda_1}{\lambda_2} \int_{\delta}^{1/2} -\phi_2'' \geq \frac{1}{1 - 2\delta} \frac{\lambda_1}{\lambda_2}. \end{aligned}$$

This completes the proof of this lemma. $\square$

In previous papers, many authors studied the bounds of the gaps and ratios of eigenvalues. In [9], the authors proved the following results for the minimum of the ratio.

Proposition 3.6 (cf. Theorem 5 in [9]) *Define the set*

$$E(M, 0, H) = \left\{ w \in L^1[0, 1] : 0 \leq w(x) \leq H, \int_0^1 w(x) dx = M \right\}.$$

Let $\lambda_1(w)$ and $\lambda_2(w)$ be the first and second eigenvalues of (1.1) and (1.2). Then

$$\frac{\lambda_2(w)}{\lambda_1(w)} \geq \left(\frac{z}{(\pi/2)^2} \right), \text{ for any } w \in E(M, 0, H),$$

where z is the least positive root of the equation $\tan z = z(1 - H/M)$.

Set

$$z_0 = \sqrt{2}\pi/2, \quad t_0 = -\tan z_0/z_0 > 0, \quad H_0 = 1 + t_0. \quad (3.12)$$

If $H/M < H_0$, then the least positive root of $\tan z = z(1 - H/M)$ is not less than z_0 . Then it follows from Proposition 3.6 that if $H \leq H_0$, then for all $w \in E(M, 0, H)$,

$$\frac{\lambda_2(w)}{\lambda_1(w)} \geq \left(\frac{z_0}{(\pi/2)^2} \right) = 2 \Rightarrow \lambda_2(w) \geq 2\lambda_1(w). \quad (3.13)$$

If $0 < c \leq w(x) \leq C$, then it is proved in [1, Theorem 4.1, (4.6)] that

$$\max\{1, 4c/C\} \leq \lambda_2(w)/\lambda_1(w) \leq 1 + 3C/c.$$

Clearly $\lambda_2(w) \geq 2\lambda_1(w)$ if $2c \geq C$.

Some authors also have studied the single-well (single-barrier) weight $w(x)$. A function in $[0, 1]$ is called single-well (single-barrier) if there is a point a in $[0, 1]$ such that it is monotone decreasing (increasing) in $[0, a]$ and monotone increasing (decreasing) in $[a, 1]$. The point a is the transition point.

Proposition 3.7 (cf. Theorems 1.1 and 1.2 in [12]) *Consider the eigenvalue problem*

$$-y''(x) = \lambda w(x)y(x), \quad y(0) = 0 = y(\pi), \quad x \in (0, \pi).$$

If $w^{-1/4}(x)$ is convex, then $\lambda_n/\lambda_m \geq [n/m]^2$, $n \geq m$. If w is symmetric single-barrier, then $\lambda_n/\lambda_1 \geq n^2$.

Since the transformation from $[0, \pi]$ to $[0, 1]$ does not change the ratios of eigenvalues, the above facts together with Theorem 3.1 immediately give the following results on the gap of eigenvalues.

Corollary 3.8 *Let λ_1 and λ_2 be the first and second eigenvalues of (1.1) and (1.2), respectively. Suppose that $w(x)$ is symmetric on $[0, 1]$ and $\int_0^1 w(x)dx = 1$.*

- (i) *If $w(x) \leq H_0$, then $\lambda_2 - \lambda_1 \geq 4\lambda_2/\lambda_1 \geq 8$, where H_0 is defined as in (3.12).*
- (ii) *If $\sup\{w(x) : x \in [0, 1]\} \leq 2 \inf\{w(x) : x \in [0, 1]\}$, then $\lambda_2 - \lambda_1 \geq 4\lambda_2/\lambda_1 \geq 8$.*
- (iii) *If $w^{-1/4}(x)$ is convex, then $\lambda_2 - \lambda_1 \geq 4\lambda_2/\lambda_1 \geq 16$.*
- (iv) *If w is symmetric single-barrier, then $\lambda_2 - \lambda_1 \geq 4\lambda_2/\lambda_1 \geq 16$.*

In [10] the authors gave the following result. If w is symmetric single-barrier, then

$$\lambda_2 - \lambda_1 \geq 3\pi^2/w(1/2),$$

see (ii) of Corollary 3.2 in [10]. We remark that the above lower bound involves the value of the weight w , so that the lower bound is implicit even though we know $\int_0^1 w(x)dx = 1$.

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